Model Predictive Control and PID for Path Following of an Unmanned Quadrotor Helicopter

Huan Cheng and Yanhua Yang
School of Information Science and Engineering
Wuhan University of Science and Technology
Wuhan, 430081, P.R. China
Email: yangyh@wust.edu.cn

***Abstract*—This paper presents a control method based on Model Predictive Control (MPC) and PID for path following of a quadrotor unmanned aerial vehicle (UAV). On the basis of simplified kinetic model, a double loop control structure is adopted. Because MPC method is ability to deal with input and output constraints, in the outer loop (position loop), MPC controllers are designed to track the reference trajectory under input or output physical constrains. In the inner loop (attitude loop), to achieve rapid response for attitude, PID controllers are designed to track the reference attitude. The simulation results validate the effectiveness of the proposed control approach.**

Keywords—quadrotor unmanned aerial vehicle (UAV); Model Predictive Control(MPC); PID; path following

I. INTRODUCTION

In recent years, the quadrotor unmanned aerial vehicle (UAV) has gradually generated great interest in the automatic control area with its simple mechanical structure and unique flight mode [1]. Quadrotor UAV is a non-coaxial and multi-rotor aircraft. The aero attitude of quadrotor is controlled by adjusting four rotors speed of the symmetrical distribution [2]. Small quadrotor UAV can carry certain equipment for autonomous flight, and its military applications, disaster relief, entertainment, scientific research and other fields have important uses and value. Moreover, it can replace the pilot to complete some dangerous rescue tasks [3]. However, the system dynamics of the quadrotor is usually strong-coupling and with nonlinearity, which makes the system difficult to be controlled in rapid and steady flights. As a result, to design a reliable controller becomes an important issue for the quadrotor control.

At present, many research institutions and universities have carried out relevant research work and put forward many control methods for the control system of the quadrotor [4], [5]. In attitude control of the quadrotor, the authors of [6] study the attitude control for a quadrotor by Ziegler-Nichols rules for tuning the PID controller method. However, tuning PID parameters is always a difficult task in the reality especially when disturbance exists. The advantage of the Ziegler-Nichols rules for tuning the PID parameters is simple and practical, but the method is not very precise. The authors of [7] study the attitude control for a quadrotor by cascade PID method. The cascade PID algorithm is efficient for attitude control which meets the response for real-time control in short time in the perturbations condition, and the disadvantage of cascade PID method is difficult to tune PID parameters. Other authors have considered adaptive sliding mode control [8], nonlinear H_∞ control [9], neural network control [10] and backstepping control [11] in the attitude control. However, these advanced control algorithms need to establish the complicate control model, which lead to time consuming in attitude control. In position control of the quadrotor, the authors of [12] study the PID and LQR control methods for path tracking of quadrotor and compare the performance of these two algorithms. Comparing with the PID controller, the tracking trajectory with LQR controller is smoother. MPC is considered to solve the autonomous flight problem of the quadrotor UAV in [13]. The authors of [14] study an explicit MPC for trajectory tracking of quadrotor. The main advantage of MPC on the flight control of a quadrotor is the fact that both the current and future physical states are taken into account, and it also can solve the problem of input and output physical constraints.

In this paper, MPC and PID control method for solving the path tracking problem of the quadrotor UAV is proposed. The objective of MPC is to compute a future control sequence in a defined horizon in such a way that the prediction of the quadrotor output is driven close to the reference. Moreover, MPC also can solve the problem of input and output constraints. The precision and speed of attitude calculation will directly affect the stability of the flight control. PID can achieve rapid response for attitude. So this paper adopts the combination of MPC and PID control method to solve the path tracking problem of the quadrotor UAV.

The remainder of the paper is organized as follows. Section Ⅱ introduces model of quadrotor UAV. Section Ⅲ describes the designed controllers. Section Ⅳ illustrates the simulation results. Finally, the conclusion is given in Section V.

II. MATHEMATICAL MODEL OF QUADROTOR

To establish dynamic model of the quadrotor, two coordinate systems are defined: the geodetic coordinate system $E(X_e, Y_e, Z_e)$ and the fuselage coordinate system $B(X_b, Y_b, Z_b)$. As shown in Fig.1, the Euler angle of the quadrotor is defined as $\Omega = [\theta, \gamma, \varphi]^T$, where θ is pitch angle, γ is roll angle, and φ is yaw angle. In this paper, it is assumed that the rotation sequence of the body is the heading yaw angle which is rotated about the Z axis, pitch angle which is rotated about the X axis, roll angle which is rotated about the Y axis, The rotation matrix of the earth

This work is supported by the National Science Foundation of China under grant 61471275 and 61573263.

Authorized licensed use limited to: DGIST. Downloaded on April 29,2026 at 18:20:41 UTC from IEEE Xplore. Restrictions apply.

coordinate system to the body coordinate system in this order is obtained:

$$R=\begin{bmatrix} \cos\varphi\cos\gamma & \sin\varphi\cos\gamma & \sin\gamma \\ \cos\varphi\sin\gamma\sin\theta-\sin\varphi\cos\theta & \sin\varphi\sin\gamma\sin\theta+\cos\varphi\cos\theta & \cos\gamma\sin\theta \\ \cos\varphi\sin\gamma\cos\theta+\sin\varphi\sin\theta & \sin\varphi\sin\gamma\cos\theta-\cos\varphi\sin\theta & \cos\gamma\cos\theta \end{bmatrix}$$

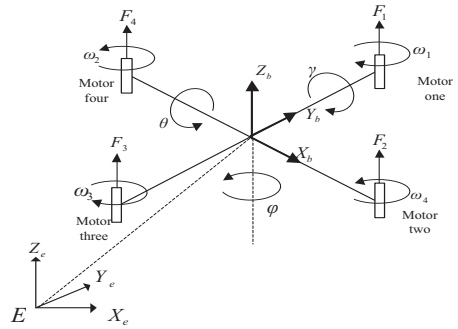

Fig. 1. Quadrotor dynamics diagram

The quadrotor motion equations can be expressed by the Newton's law of motion and Euler's equation[15]:

$$\boldsymbol{F}=m\ddot{\boldsymbol{P}} \tag{1}$$

$$\boldsymbol{M}=\frac{d\boldsymbol{H}}{dt} \tag{2}$$

Where $\boldsymbol{F}$ is resultant external force acting on quadrotor; m is the mass of quadrotor; $\boldsymbol{M}$ is rotational torque of quadrotor; $\boldsymbol{H}$ is the angular momentum of the body under the navigation system; $\boldsymbol{P}=[x \quad y \quad z]^T$ is the center of mass position of body under the navigation system; b and d represent lift coefficient and drag coefficient respectively; l is the distance between each propeller and the center of gravity of the quadrotor; J_x 、 J_y 、 J_z are the moment of inertia with respect to x, y and z axis respectively; $\omega_i(i=1,2,3,4)$ is the rotational speed of the ith rotor. $\boldsymbol{F}=[F_{Ex} \quad F_{Ey} \quad F_{Ez}]^T$ is external force that the quadrotor is subjected in the navigation system; $\boldsymbol{M}_E=[M_{Ex} \quad M_{Ey} \quad M_{Ez}]^T$ is torque that body is subjected. Quadrotor UAV in the absence of wind and slow flight conditions, the air resistance can be negligible. After a series of derivation and simplification, the dynamic model of the quadrotor in the navigation system is obtained:

$$\begin{cases} F_{Ex}=(\cos\phi\sin\theta\cos\varphi+\sin\phi\sin\varphi)\cdot b(\omega_1^2+\omega_2^2+\omega_3^2+\omega_4^2) \\ F_{Ey}=(\sin\varphi\sin\theta\cos\phi-\cos\varphi\sin\phi)\cdot b(\omega_1^2+\omega_2^2+\omega_3^2+\omega_4^2) \\ F_{Ez}=(\cos\phi\cos\theta)\cdot b(\omega_1^2+\omega_2^2+\omega_3^2+\omega_4^2)-g \\ M_{Ex}=-(J_y-J_z)\dot{\theta}\dot{\varphi}+bl(\omega_1^2-\omega_3^2) \\ M_{Ey}=-(J_z-J_x)\dot{\phi}\dot{\varphi}+bl(\omega_2^2-\omega_4^2) \\ M_{Ez}=-(J_x-J_y)\dot{\theta}\dot{\phi}+d(\omega_1^2+\omega_3^2-\omega_2^2-\omega_4^2) \end{cases} \tag{3}$$

Equation (3) decouples the quadrotor in six degrees of freedom, quadrotor UAV flight status is divided into four independent channels: up and down, left and right, front and rear, yaw, the definition of variables:

$$\begin{cases} U_1=b(\omega_1^2+\omega_2^2+\omega_3^2+\omega_4^2) \\ U_2=b(\omega_1^2-\omega_3^2) \\ U_3=b(\omega_2^2-\omega_4^2) \\ U_4=d(\omega_1^2+\omega_3^2-\omega_2^2-\omega_4^2) \end{cases} \tag{4}$$

U_1 is the upper and lower channel control; U_2 is the front and rear channel and pitch angle control volume; U_3 is the left and right channel and roll angle control volume; U_4 is the yaw angle control quantity. According to (3) and (4), equation (5) can be obtained:

$$\begin{cases} \ddot{x}=(\cos\varphi\sin\gamma\cos\theta+\sin\varphi\sin\theta)\cdot\dfrac{U_1}{m} \\ \ddot{y}=(\sin\varphi\sin\gamma\cos\theta-\cos\varphi\sin\theta)\cdot\dfrac{U_1}{m} \\ \ddot{z}=(\cos\gamma\cos\theta)\cdot\dfrac{U_1}{m}-g \\ \ddot{\theta}=\dfrac{-(J_y-J_z)}{J_x}\dot{\gamma}\dot{\varphi}+\dfrac{lU_2}{J_x} \\ \ddot{\gamma}=\dfrac{-(J_z-J_x)}{J_y}\dot{\theta}\dot{\varphi}+\dfrac{lU_3}{J_y} \\ \ddot{\varphi}=\dfrac{-(J_x-J_y)}{J_z}\dot{\theta}\dot{\gamma}+\dfrac{U_4}{J_z} \end{cases} \tag{5}$$

III. Control System Design

In order to fulfill path tracking control of the quadrotor UAV in the flight, the control structure is shown in Fig.2. The attitude control is taken as the inner loop, the position control as the outer loop, and the attitude loop is nested in the position loop. $\boldsymbol{P}_r=[x_r \quad y_r \quad z_r]$ is the reference path vector that the trajectory generator inputs, and $\boldsymbol{U}_r=[U_{1r} \quad u_{xr} \quad u_{yr}]$ is the control vector calculated by MPC algorithm. $\boldsymbol{R}_r=[\theta_r \quad \gamma_r \quad \varphi_r]$ is reference angles vector calculated by $\boldsymbol{U}_r$.

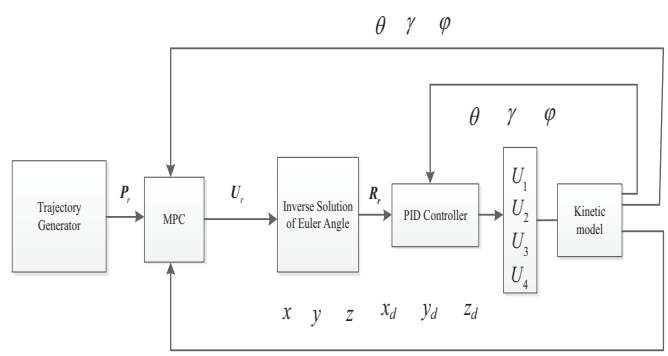

Fig.2. Control structure diagram

A. Model Predictive Controller(MPC) for Position Control

Due to MPC is the fact that both the current and future physical states are taken into account, and moreover it can cope with the input and output constraints, this paper adopts MPC to achieve position control of quadrotor UAV. Position control is divided into height control and x, y direction of control. Height

Authorized licensed use limited to: DGIST. Downloaded on April 29,2026 at 18:20:41 UTC from IEEE Xplore. Restrictions apply.

controller calculates the input U_1, and the controllers in the x and y directions calculate the control inputs u_x, u_y by U_1. Therefore, U_1 is obtained firstly. A linear state space MPC method is adopted based on the quadrotor dynamic model. The system (5) is rewritten as a state-space form $\dot{\bar{\xi}} = f(\bar{\xi}(t), \boldsymbol{u}_\xi(t))$ for the controller, where $\bar{\xi}(t) = [x(t)\ u_0(t)\ y(t)\ v_0(t)\ z(t)\ w_0(t)]'$ stands for the state-space vector of the system [1]. From (5) and the new state-space vector, the system dynamic equation to control design can be written in the following form:

$$\dot{\bar{\xi}} = f(\bar{\xi}(t), \boldsymbol{u}_\xi(t)) = \begin{bmatrix} u_0 \\ u_x(t)\dfrac{U_1(t)}{m} \\ v_0 \\ u_y(t)\dfrac{U_1(t)}{m} \\ w_0 \\ -g + (\cos\theta(t)\cos\gamma(t))\dfrac{U_1(t)}{m} \end{bmatrix} \tag{6}$$

with:

$$\begin{aligned} u_x(t) &\triangleq \cos\varphi(t)\sin\gamma(t)\cos\theta(t) + \sin\varphi(t)\sin\theta(t) \\ u_y(t) &\triangleq \sin\varphi(t)\sin\gamma(t)\cos\theta(t) - \cos\varphi(t)\sin\theta(t) \end{aligned} \tag{7}$$

Due to the fact that the target coordinates vary in time, in order to make error minimized, a virtual reference vehicle with the same quadrotor mathematical model is defined:

$$\dot{\bar{\xi}}_r = f(\bar{\xi}_r(t), \boldsymbol{u}_{\xi r}(t)) \tag{8}$$

where $\bar{\xi}_r(t) = [x_r(t)\ u_{0r}(t)\ y_r(t)\ v_{0r}(t)\ z_r(t)\ w_{0r}(t)]'$ and $\boldsymbol{u}_{\xi r}(t) = [u_{xr}\ u_{yr}\ U_{1r}]'$ are the reference states and the control inputs respectively. Null external disturbances are assumed in the virtual reference vehicle, the reference values are given by:

$$U_{1r} = m\cdot(\ddot{z}_r + g),\ u_{xr} = \frac{\ddot{x}_r \cdot m}{U_{1r}},\ u_{yr} = \frac{\ddot{y}_r m}{U_{1r}}$$

According to (6) and (8), the proposed translational error model is given by:

$$\dot{\tilde{\bar{\xi}}}(t) = \boldsymbol{A}(t)\cdot\tilde{\bar{\xi}}(t) + \boldsymbol{B}(t)\cdot\tilde{\boldsymbol{u}}_\xi(t) \tag{9}$$

where $\tilde{\bar{\xi}}(t) = \bar{\xi}(t) - \bar{\xi}_r(t)$ represents the error vector, and $\tilde{\boldsymbol{u}}_\xi(t) = \boldsymbol{u}_\xi(t) - \boldsymbol{u}_{\xi r}(t)$ is the control input error. Matrices ***A(t)*** and ***B(t)*** are the Jacobian for the system (6) in relation to $\bar{\xi}(t)$ and $u_\xi(t)$. Besides, integral of the position error term has been included in the error vector to perform an appropriate path following, in order to obtain a better tracking trajectory. The following augmented error vector is considered:

$$\boldsymbol{x}_\xi(t) = \begin{bmatrix} \tilde{x}(t) \\ \tilde{u}_0(t) \\ \int \tilde{x}(t)dt \\ \tilde{y}(t) \\ \tilde{v}_0 \\ \int \tilde{y}(t)dt \\ \tilde{z}(t) \\ \tilde{w}_0(t) \\ \int \tilde{z}(t)dt \end{bmatrix} = \begin{bmatrix} x(t) - x_r(t) \\ u_0(t) - u_{0r}(t) \\ \int (x(t) - x_r(t))dt \\ y(t) - y_r(t) \\ v_0(t) - v_{0r}(t) \\ \int (y(t) - y_r(t))dt \\ z(t) - z_r(t) \\ w_0(t) - w_{0r}(t) \\ \int (z(t) - z_r(t))dt \end{bmatrix} \tag{10}$$

A time varying discrete linear equation (11) can be obtained by discretization the differential (6) based Euler scheme.

$$\boldsymbol{x}_\xi(k+1) = \bar{\boldsymbol{A}}(k)\cdot\boldsymbol{x}_\xi(k) + \bar{\boldsymbol{B}}(k)\cdot\tilde{\boldsymbol{u}}_\xi(k) \tag{11}$$

The error model (11) can be divided into two subsystems. Height error can take the following form:

$$\boldsymbol{x}_{\xi z}(k+1) = \bar{\boldsymbol{A}}_z(k)\cdot\boldsymbol{x}_{\xi z}(k) + \bar{\boldsymbol{B}}_z(k)\cdot\tilde{\boldsymbol{u}}_{\xi z}(k) \tag{12}$$

Matrices $\bar{\boldsymbol{A}}_z(t)$ and $\bar{\boldsymbol{B}}_z(t)$ for height error are the following:

$$\bar{\boldsymbol{A}}_z = \begin{bmatrix} 1 & \Delta t & 0 \\ 0 & 1 & 0 \\ \Delta t & 0 & 1 \end{bmatrix},\ \bar{\boldsymbol{B}}_z = \begin{bmatrix} 0 \\ \dfrac{\Delta t}{m}\cos\theta(k)\cos\phi(k) \\ 0 \end{bmatrix} \tag{13}$$

The x and y motions error can take the following form:

$$\boldsymbol{x}_{\xi xy}(k+1) = \bar{\boldsymbol{A}}_{xy}(k)\cdot\boldsymbol{x}_{\xi xy}(k) + \bar{\boldsymbol{B}}_{xy}(k)\cdot\tilde{\boldsymbol{u}}_{\xi xy}(k) \tag{14}$$

Matrices $\bar{\boldsymbol{A}}_{xy}(t)$ and $\bar{\boldsymbol{B}}_{xy}(t)$ for the x and y motions are the following:

$$\bar{\boldsymbol{A}}_{xy} = \begin{bmatrix} 1 & \Delta t & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ \Delta t & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & \Delta t & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \Delta t & 0 & 1 \end{bmatrix} \tag{15}$$

$$\bar{\boldsymbol{B}}_{xy} = \begin{bmatrix} 0 & 0 \\ \dfrac{\Delta t}{m}U_1(k) & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & \dfrac{\Delta t}{m}U_1(k) \\ 0 & 0 \end{bmatrix} \tag{16}$$

where Δt is the sampling time.

Authorized licensed use limited to: DGIST. Downloaded on April 29,2026 at 18:20:41 UTC from IEEE Xplore. Restrictions apply.

The $\bar{A}_{xy}(t)$ matrix in (15) is constant in each time, and the $\bar{B}_{xy}(t)$ matrix in (16) is changing for the U_1, which is considered as a time-varying parameter in each time. Moreover, because of the decentralized control structure, the roll, pitch and yaw angles are also considered as time-varying parameters.

According to the above analysis, quadrotor UAV path tracking is to find a suitable control input, minimizing the tracking error. Therefore the path tracking problem is converted to solve QP problems with input constrains. For the height control, the control input U_1 is calculated by optimizing the following QP problem.

$$\begin{aligned}\min J_z &= \left[\hat{\boldsymbol{x}}_{\xi z} - \hat{\boldsymbol{x}}_{\xi rz}\right]' \boldsymbol{Q}_z \left[\hat{\boldsymbol{x}}_{\xi z} - \hat{\boldsymbol{x}}_{\xi rz}\right] \\ &+ \left[\hat{\tilde{\boldsymbol{u}}}_{\xi z} - \hat{\tilde{\boldsymbol{u}}}_{\xi rz}\right]' \boldsymbol{R}_z \left[\hat{\tilde{\boldsymbol{u}}}_{\xi z} - \hat{\tilde{\boldsymbol{u}}}_{\xi rz}\right]\end{aligned} \tag{17}$$

Subject to:

$$\hat{\boldsymbol{x}}_{\xi z} - \boldsymbol{P}_z(k|k)\cdot \boldsymbol{x}_{\xi z}(k|k) - \boldsymbol{H}_z(k|k)\cdot \hat{\tilde{\boldsymbol{u}}}_{\xi z} = 0$$

$$\hat{\tilde{u}}_{\xi z} < u_{z\max} \quad (i = 1, 2\cdots N_u)$$

where $\boldsymbol{Q}_z$ and R_z are diagonal definite positive weighting matrices. The predictions of the model output $\hat{\boldsymbol{x}}_{\xi z}(k+j|k)$ are computed using a linearized time-varying state-space model of the vehicle by (12) and (13), obtaining:

$$\hat{\boldsymbol{x}}_{\xi z} = \boldsymbol{P}_z(k|k)\cdot \boldsymbol{x}_{\xi z}(k|k) + \boldsymbol{H}_z(k|k)\cdot \hat{\tilde{\boldsymbol{u}}}_{\xi z} \tag{18}$$

where $\tilde{\boldsymbol{u}}_{\xi z}(k|k) = U_1(k) - U_{1r}(k)$ and $\boldsymbol{x}_{\xi z}(k)$ is the height state error vector. The height state and control input reference vectors are:

$$\hat{\boldsymbol{x}}_{\xi rz} \triangleq \begin{bmatrix} \boldsymbol{x}_{\xi rz}(k+1|k) - \boldsymbol{x}_{\xi rz}(k|k) \\ \vdots \\ \boldsymbol{x}_{\xi rz}(k+N_z|k) - \boldsymbol{x}_{\xi rz}(k|k) \end{bmatrix}$$

$$\hat{\tilde{\boldsymbol{u}}}_{\xi rz} \triangleq \begin{bmatrix} U_{1r}(k|k) - U_{1r}(k-1|k) \\ \vdots \\ U_{1r}(k+N_{uz}-1|k) - U_{1r}(k-1|k) \end{bmatrix}$$

where N_z is the prediction horizon and N_{uz} is the control horizon. At each time k, $\hat{\tilde{\boldsymbol{u}}}_{\xi z}(k|k)$ can be calculated by (17). Therefore, the following control signal is applied to the quadrotor:

$$U_1(k) = \hat{\tilde{\boldsymbol{u}}}_{\xi z}(k|k) + U_{1r}(k)$$

The second control law computes the x and y motion control inputs, and the same previous procedure is carried out using the error model (14), (15) and (16). The optimization problem is expressed as follows.

$$\begin{aligned}\min J_{xy} &= \left[\hat{\boldsymbol{x}}_{\xi xy} - \hat{\boldsymbol{x}}_{\xi rxy}\right]' \boldsymbol{Q}_{xy} \left[\hat{\boldsymbol{x}}_{\xi xy} - \hat{\boldsymbol{x}}_{\xi rxy}\right] \\ &+ \left[\hat{\tilde{\boldsymbol{u}}}_{\xi xy} - \hat{\tilde{\boldsymbol{u}}}_{\xi rxy}\right]' \boldsymbol{R}_z \left[\hat{\tilde{\boldsymbol{u}}}_{\xi xy} - \hat{\tilde{\boldsymbol{u}}}_{\xi rxy}\right]\end{aligned} \tag{19}$$

Subject to:

$$\hat{\boldsymbol{x}}_{\xi xy} - \boldsymbol{P}_{xy}(k|k)\cdot \boldsymbol{x}_{\xi xy}(k|k) - \boldsymbol{H}_{xy}(k|k)\cdot \hat{\tilde{\boldsymbol{u}}}_{\xi xy} = 0$$

$$\hat{\tilde{\boldsymbol{u}}}_{\xi xy}(i) < u_{xy\max} \quad (i = 1, 2\cdots N_u)$$

where

$\hat{\tilde{\boldsymbol{u}}}_{\xi xy}(k|k) = \left[\tilde{u}_x(k|k)\ \tilde{u}_y(k|k)\right]'$, $\boldsymbol{u}_{\xi xy}(k) = \boldsymbol{u}_{\xi rxy}(k) + \hat{\tilde{\boldsymbol{u}}}_{\xi xy}(k|k)$. The reference vectors of the error states, $\hat{\boldsymbol{x}}_{\xi rxy}$, and the error control inputs, $\hat{\tilde{\boldsymbol{u}}}_{\xi rxy}$, are obtained by the same way as the one of the height controller case.

B. PID for Attitude Control

The attitude control is the key to control of quadrotor. Because PID provides a good response for the model dynamics attitudes, this paper uses PID control method for the attitude of the quadrotor UAV. U_{1r}, u_{xr}, u_{yr} are calculated by position control. φ_r is obtained at each moment by feedback of the kinetic equation. Equation (7) can be obtained:

$$\theta_r = \arcsin(u_{xr}\sin(\varphi_r) - u_{yr}\cos(\varphi_r))$$

$$\gamma_r = \arcsin\left(\frac{u_{xr}\cos(\varphi_r) + u_{yr}\sin(\varphi_r)}{\cos(\theta_r)}\right)$$

PID is defined:

$$\begin{cases} e_\theta = K_{P\theta}(\theta_r - \theta) + K_{I\theta}\int(\theta_r - \theta)dt + K_{D\theta}(\dot{\theta}_r - \dot{\theta}) \\ e_\gamma = K_{P\gamma}(\gamma_r - \gamma) + K_{I\gamma}\int(\gamma_r - \gamma)dt + K_{D\gamma}(\dot{\gamma}_r - \dot{\gamma}) \\ e_\varphi = K_{P\varphi}(\varphi_r - \varphi) + K_{I\varphi}\int(\varphi_r - \varphi)dt + K_{D\varphi}(\dot{\varphi}_r - \dot{\varphi}) \end{cases} \tag{20}$$

According to (4) and (20), quadrotor UAV flight rotation characteristics are obtained:

$$\begin{cases} \omega_1^2 = e_\theta + e_\varphi + \dfrac{U_1}{4b} \\ \omega_2^2 = e_\gamma - e_\varphi + \dfrac{U_1}{4b} \\ \omega_3^2 = -e_\theta + e_\varphi + \dfrac{U_1}{4b} \\ \omega_4^2 = -e_\gamma - e_\varphi + \dfrac{U_1}{4b} \end{cases} \tag{21}$$

U_2, U_3, U_4 is obtained by (21) and (4).

IV. Simulation Results and Analysis

In this section, we have carried out series of simulations to check the performance of the quadrotor in following the reference trajectory by using MPC and PID methods.

The weighted matrices Q, R and control constraints $u_{z\max}$, $u_{xy\max}$ for the implementation of the MPC are given as follows:

$$Q_z = diag(100, 4, 17),\ R_z = 0.01,\ u_{z\max} = 30$$

Authorized licensed use limited to: DGIST. Downloaded on April 29,2026 at 18:20:41 UTC from IEEE Xplore. Restrictions apply.

$$Q_{xy} = diag(2.4, 0.1, 0.8, 2.4, 0.1, 0.8),\ R_{xy} = diag(0.1, 0.1),$$
$$u_{xy\max} = 10$$

The gains for the implementation of the PID control are listed as follows.

$$K_{p1} = 0.5, K_{I1} = 0.01, K_{D1} = 3$$
$$K_{p2} = 1, K_{I2} = 0.01, K_{D2} = 0.5$$
$$K_{p3} = 0.05, K_{I3} = 0.01, K_{D3} = 0.5$$

The input reference path is chosen as the same as [1]:

$$x_r = \frac{1}{2}\cos(\frac{\pi t}{20}), y_r = \frac{1}{2}\sin(\frac{\pi t}{20}), z_r = 3 - 2\cos(\frac{\pi t}{20})$$

A good tracking of the reference trajectory for quadrotor UAV flight is showed in Fig.3 and Fig.4. Fig.3 is a three-dimensional trajectory tracking, and Fig.4 shows the path tracking in x, y and z directions, respectively. As displayed in the Fig.3 and Fig.4, quadrotor UAV can achieve good path tracking in the flight by using MPC and PID methods.

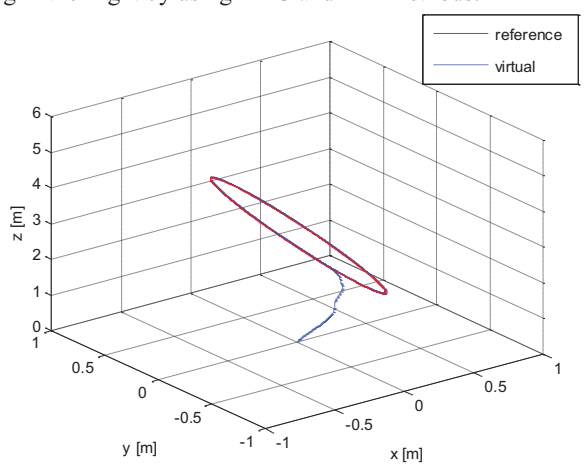

Fig.3. Quadrotor three dimensional path tracking diagram

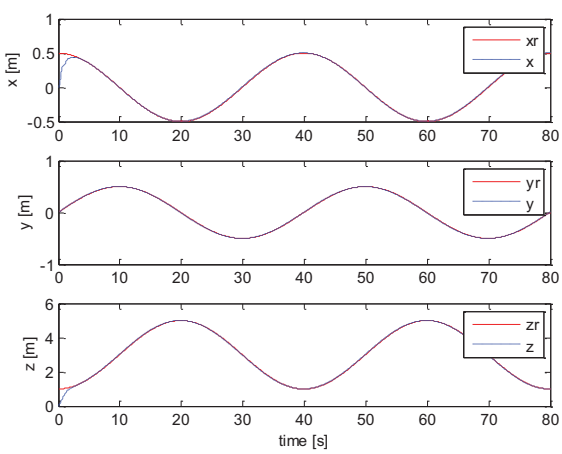

Fig.4. Position following (x, y, z)

Fig.5 shows the translational coordinates errors. It can be seen that translational coordinates errors almost is zero. As displayed in the Fig.6, it can be seen that the attitude value quickly stabilized to the set point by using PID methods.

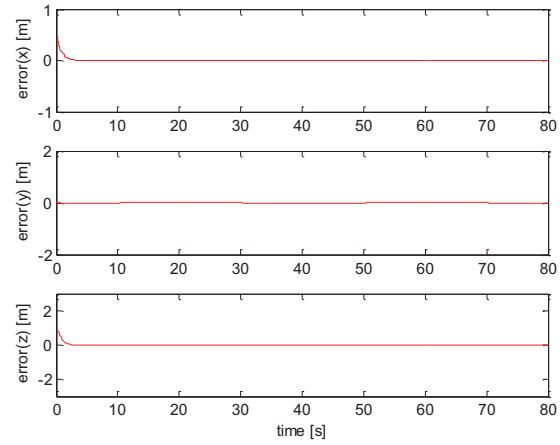

Fig.5.Position error (x, y, z)

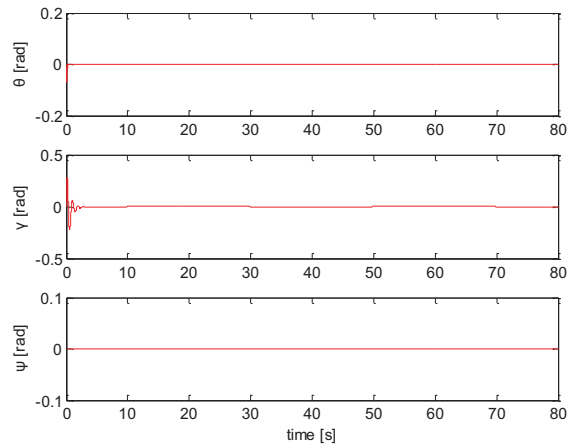

Fig.6. Orientation $(\theta, \gamma, \varphi)$

Based on the above analysis, the model predictive control and PID control can realize the path tracking in the quadrotor UAV flight.

In order to further illustrate the effectiveness of the proposed method, the method of this paper is compared with the double PID scheme that the inner loop and outer loop both adopt PID algorithm. The comparative results are shown in Fig.7. Fig.7 shows PID control value and MPC control value respectively. The max represents maximum control constraint, the UP resprents control value of PID, the UM resprents control value of MPC. It can be seen that control value of double PID scheme exceeds the max in some moment. The control value U1 of our method does not exceed the max at all time. So, the proposed control method in this paper has an advantage over double PID method.

Authorized licensed use limited to: DGIST. Downloaded on April 29,2026 at 18:20:41 UTC from IEEE Xplore. Restrictions apply.

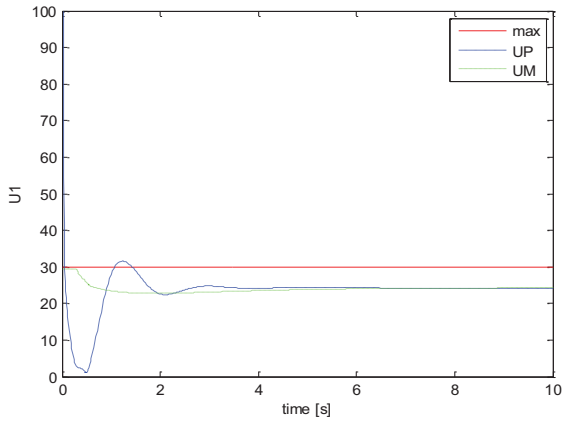

Fig.7.Compare U1 of PID and MPC

V. CONCLUSIONS

In this paper, Model Prediction Control (MPC) and PID control strategy is proposed to solve the path tracking problem for a quadrotor (UAV). The objective of MPC is to compute a future control sequence in a defined horizon in such a way that the prediction of the quadrotor output is driven close to the reference, and further MPC method can deal with input and output constraints. In order to achieve rapid response for attitude, we designed PID controllers to track the reference attitude. Finally, the simulation results verify the effectiveness of our proposed method.

REFERENCES

[1] Raffo G V, Ortega M G, Rubio F R. An integral predictive/nonlinear H_∞ control structure for a quadrotor helicopter, Automatica, 2010, 46(1): 29-39.

[2] Li L B, Sun L L, Jin J. Survey of advances in control algorithms of quadrotor unmanned aerial vehicle, IEEE 16th International Conference on Communication Technology. 2015: 107-111.

[3] Abdolhosseini M, Zhang Y M, Rabbath C A. An efficient model predictive control scheme for an unmanned quadrotor helicopter, Journal of Intelligent & Robotic Systems, 2013, 70(1): 27-38.

[4] Hoffmann G M, Huang H, Waslander S L, Tomlin C J. Precision flight control for a multi-vehicle quadrotor helicopter testbed, Control Engineering Practice, 2011, 19(9): 1023-1036.

[5] Alexis K, Nikolakopoulos G, Tzes A. Switching model predictive attitude control for a quadrotor helicopter subject to atmospheric disturbances, Control Engineering Practice, 2011, 19(10): 1195-1207.

[6] He Z F, Zhao L. A simple attitude control of quadrotor helicopter based on Ziegler-Nichols rules for tuning PD parameters. The scientific world journal, 2014.

[7] Gao P B, Liu Y X, Zhang H, Wang L L. Quadrotor helicopter attitude control using cascade PID, Chinese Control and Decision Conference (CCDC), 2016: 5158-5163.

[8] Islam S, Faraz M, Ashour R K, Cai G, Dias J, Seneviratne L. Adaptive sliding mode control design for quadrotor unmanned aerial vehicle, International Conference on Unmanned Aircraft Systems (ICUAS), 2015: 34-39.

[9] Jasim W, Gu D. H_∞ for quadrotor attitude stabilization. UKACC International Conference on Control (CONTROL), 2014: 19-24.

[10] Dierks T, Jagannathan S. Output feedback control of a quadrotor UAV using neural networks, IEEE transactions on neural networks, 2009, 21(1): 50-66.

[11] Chingozha T, Nyandoro O. Adaptive sliding backstepping control of quadrotor UAV attitude, IFAC Proceedings Volumes, 2014, 47(3):11043-11048.

[12] Liu C L, Pan J, Chang Y F. PID and LQR trajectory tracking control for an unmanned quadrotor helicopter: Experimental studies, 35th Chinese Control Conference, 2016: 10845-10850.

[13] Alexis K, Nikolakopoulos G, Tzes A. Model predictive control scheme for the autonomous flight of an unmanned quadrotor, IEEE International Symposium on Industrial Electronics, 2011: 2243-2248.

[14] Liu C J, Lu H, Chen W H. An explicit MPC for quadrotor trajectory tracking, 34th Chinese Control Conference. 2015: 4055-4060.

[15] Bai Y Q, Liu H, Shi Z Y, Robust flight control of a four-rotor unmanned helicopter, Robotics, 2012, 34 (5): 9-14.

Authorized licensed use limited to: DGIST. Downloaded on April 29,2026 at 18:20:41 UTC from IEEE Xplore. Restrictions apply.